Assignment 7B. RK4 – 1st & 2nd order ODE Marks 20

Posted on 14.10.2026 @ 2:30 pm and due on 14.10.2026 @ 5:40 pm

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1. Write library routine for Runge-Kutta-4. Make sure you can call / use the same RK4 subroutine for both first and second order ODE. **[5]**

2. Use Runge-Kutta-4 method to solve and the following differential equation and compare with analytical form over the range $x \in [0, \pi/5]$. Use step sizes 0.05, 0.15. **[5]**

$$\frac{dy}{dx} = (x+y)^2 \quad \text{with} \;\; y(0) = 1 \quad \rightarrow \quad \tan^{-1}(x+y) = x + \frac{\pi}{4}$$

3. Use RK4 to solve the damped simple harmonic oscillator using the following initial values and parameters over a time range $[0, 40]$. Plot variation of x, v and total energy E with time t. **[5]**

$\ddot{x} + \mu\,\dot{x} + \omega^2 x = 0$, with $x(0) = 1.0$, $v(0) = 0.0$ and $k = 1.0$, $m = 1.0$, $\mu = 0.15$

4. Solve the Rössler system for $a = 0.2$, $b = 0.2$, $c = 5.7$ and show a 3-dim (x, y, z) plot. **[5]**

$$\frac{dx}{dt} = -y - z, \;\; \frac{dy}{dt} = x + a\,y, \;\; \frac{dz}{dt} = b + z\,(x - c)$$